

# Calculus II: 2024 fall

1 a)  $\int_0^\pi \int_x^\pi \frac{\sin y}{y} dy dx$ .

Sol<sup>n</sup>,

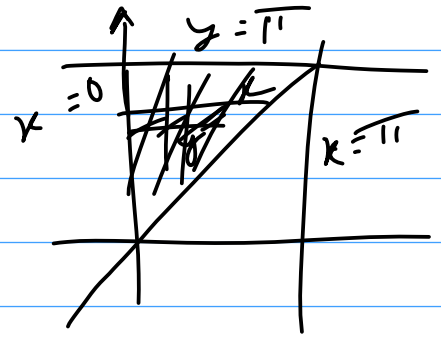
Region:  $x \leq y \leq \pi$  ;  $y=x$  ,  $y=\pi$

$0 \leq x \leq \pi$  ;  $x=0$  ,  $x=\pi$

Changing order of integration,

$$\int_0^\pi \int_0^y \frac{\sin y}{y} dy dx$$

$$= \int_0^\pi \frac{\sin y}{y} \cdot y dy \Rightarrow \int_0^\pi \sin y dy \Rightarrow [-\cos y]_0^\pi = 1 + 1 = \underline{\underline{2}}$$



1.b)  $\iiint_V (x^2 + y^2 + z^2) dx dy dz$  , volume of sphere:  $x^2 + y^2 + z^2 = 1$

Sol<sup>n</sup>,

Region:  $z = \pm \sqrt{1 - x^2 - y^2}$

Put  $z=0$  ;  $y = \pm \sqrt{1 - x^2}$

Put  $z=0$  ,  $y=0$  ;  $x = \pm 1$

Change to polar,

$x = r \cos \phi \cos \theta$  ,  $y = r \sin \phi \cos \theta$  ,  $z = r \sin \theta$  , so that  $x^2 + y^2 + z^2 = r^2$

$dx dy dz = r^2 \sin \theta dr d\theta d\phi$  ;  $0 \leq r \leq 1$  ,  $0 \leq \theta \leq \frac{\pi}{2}$  ,  $0 \leq \phi \leq \frac{\pi}{2}$   
for 1<sup>st</sup> octant.

$$\iiint_V (x^2 + y^2 + z^2) dx dy dz = 8 \int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} \int_0^1 r^2 \cdot r^2 \sin \theta dr d\theta d\phi$$

$$= 8 \int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} \sin \theta \left[ \frac{r^5}{5} \right]_0^1 d\theta d\phi$$

$$= \frac{8}{5} \int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} \sin \theta d\theta d\phi \Rightarrow \frac{8}{5} \int_0^{\frac{\pi}{2}} \sin \theta \cdot \frac{\pi}{2} d\theta \Rightarrow \frac{4\pi}{5} \int_0^{\frac{\pi}{2}} \sin \theta d\theta$$

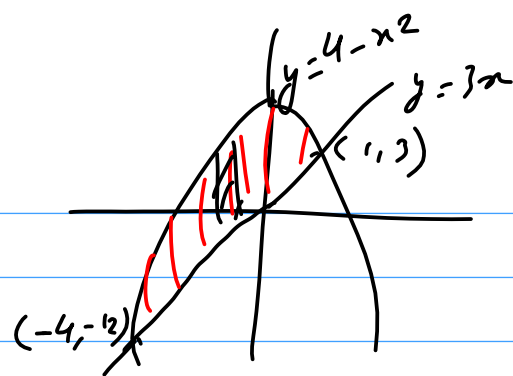
$$\Rightarrow \frac{4\pi}{5} [-\cos \theta]_0^{\frac{\pi}{2}} = \frac{4\pi}{5} //$$

1.c)  $z = x + 4$  ;  $y = 4 - x^2$ ,  $y = 3x$

$$y = 4 - x^2 \quad | \quad y = 3x$$

|     |   |         |         |
|-----|---|---------|---------|
| $x$ | 0 | $\pm 2$ | $\pm 3$ |
| $y$ | 4 | 0       | -5      |

|     |   |   |
|-----|---|---|
| $x$ | 0 | 1 |
| $y$ | 0 | 3 |



Solve

$$3x = 4 - x^2$$

$$x^2 + 3x - 4 = 0$$

$$x = 1, -4 \quad (1, 3)$$

$$y = 3, -12 \quad (-4, -12)$$

Region 1  $3x \leq y \leq 4 - x^2$   
 $-4 \leq x \leq 1$

$$\text{Reqd volume} = \iint_R z \, dA = \int_{-4}^1 \int_{3x}^{4-x^2} (x+4) \, dy \, dx$$

$$\Rightarrow \int_{-4}^1 (x+4) \left( y \right)_{3x}^{4-x^2} dx$$

$$\Rightarrow \int_{-4}^1 (x+4) (4 - x^2 - 3x) dx$$

$$\Rightarrow \int_{-4}^1 (4x - x^3 - 3x^2 + 16 - 4x^2 - 12x) dx$$

$$\Rightarrow \int_{-4}^1 (-x^3 - 7x^2 - 8x + 16) dx$$

$$= \left[ -\frac{x^4}{4} - \frac{7x^3}{3} - \frac{8x^2}{2} + 16x \right]_{-4}^1$$

$$= \left[ -\frac{1}{4} - \frac{7}{3} - \frac{8}{2} + 16 - \left( \frac{(-4)^4}{4} - \frac{7(-4)^3}{3} - \frac{8(-4)^2}{2} + 16(-4) \right) \right]$$

$$\Rightarrow \frac{625}{12} //$$

2.a) Solve using power series;  $y'' + 9y = 0$ .  $\rightarrow$  ①

Sol,

Let,  $y = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots$   $\rightarrow$  ②

$$y' = \sum_{n=1}^{\infty} n a_n x^{n-1}, \quad y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}$$

Put  $y''$ ,  $y$  in ①,

$$\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + 9 \sum_{n=0}^{\infty} a_n x^n = 0$$

Put  $k = n-2$  & replace with  $n$ ,

$$\sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2} x^n + 9 \sum_{n=0}^{\infty} a_n x^n = 0$$

$$(n+1)(n+2)a_{n+2} + 9a_n = 0$$

$$a_{n+2} = \frac{-9a_n}{(n+1)(n+2)}$$

Put  $n=0$ , Put  $n=1$ , Put  $n=2$ , Put  $n=3$ ,  
 $a_2 = \frac{-9a_0}{2!}$ ,  $a_3 = \frac{-9a_1}{3! \times 3}$ ,  $a_4 = \frac{-9a_2}{4!}$ ,  $a_5 = \frac{-9a_3}{5! \times 3}$

Put  $n=4$ , Put  $n=5$ ,  
 $a_6 = \frac{-9a_4}{6!}$ ,  $a_7 = \frac{-9a_5}{7! \times 3}$

Substitute coefficients in (2),  
 $y = a_0 + a_1 x - \frac{3^2 a_0}{2!} x^2 - \frac{3^3 a_1}{3! \times 3} x^3 + \frac{3^4 a_0}{4!} x^4 + \frac{3^5 a_1}{5! \times 3} x^5 - \frac{3^6 a_0}{6!} x^6 - \frac{3^7 a_1}{7! \times 3} x^7 + \dots$

$$\therefore y = a_0 \left( 1 - \frac{3^2}{2!} x^2 + \frac{3^4}{4!} x^4 - \frac{3^6}{6!} x^6 + \dots \right) + a_1 \left( x - \frac{3^3}{3! \times 3} x^3 + \frac{3^5}{5! \times 3} x^5 - \frac{3^7}{7! \times 3} x^7 + \dots \right)$$

$$y = a_0 \cos 3x + \frac{a_1}{3} \sin 3x //$$

2.b) solution of Legendre's eqn.

3.a)  $y'' + 2y' - 3y = 6e^{-2t}$ ,  $y(0) = 2$ ,  $y'(0) = -14$

$$\Rightarrow L(y'' + 2y' - 3y) = L(6e^{-2t})$$

$$\text{or, } [s^2 L(y) - s y(0) - y'(0)] + 2[s L(y) - y(0)] - 3L(y) = \frac{6}{s+2}$$

$$\text{or, } L(y)[s^2 + 2s - 3] - 2s + 14 - 4 = \frac{6}{s+2}$$

$$\text{or, } L(y)(s-1)(s+3) = \frac{6}{s+2} + 2s - 10$$

$$\text{or, } L(y)(s-1)(s+3) = \frac{6 + 2s^2 + 4s - 10s - 20}{s+2} \Rightarrow \frac{2s^2 - 6s - 14}{(s-1)(s+2)(s+7)}$$

$$y = L^{-1} \left[ \frac{2s^2 - 6s - 14}{(s-1)(s+2)(s+3)} \right]$$

$$\text{Or, } \frac{2s^2 - 6s - 14}{(s-1)(s+2)(s+3)} = \frac{A}{(s-1)} + \frac{B}{(s+2)} + \frac{C}{(s+3)}$$

$$\text{Or, } 2s^2 - 6s - 14 = A(s+2)(s+3) + B(s-1)(s+3) + C(s-1)(s+2)$$

$$\text{Put } s=1, \quad \text{Put } s=-2, \quad \text{Put } s=-3$$

$$\text{Or, } -18 = 12A \quad 6 = -3B \quad 22 = 4C$$

$$\therefore A = -\frac{3}{2} \quad \therefore B = -2 \quad \therefore C = \frac{11}{2}$$

s.,

$$y = L^{-1} \left[ \frac{-3}{2(s-1)} - \frac{2}{(s+2)} + \frac{11}{2(s+3)} \right]$$

$$\therefore y = -\frac{3}{2} e^t - 2e^{-2t} + \frac{11}{2} e^{-3t} //$$

3.b) 2<sup>nd</sup> shifting property of Laplace transform.

i) If  $L(f(t)) = F(s)$  then,

$$L(f(t-a)u(t-a)) = e^{-as} F(s)$$

$$\text{Also, } L(f(t)u(t-a)) = e^{-as} L(f(t+a))$$

$$f(t) = e^{-3t} u(t-3)$$

$$\begin{aligned} L(e^{-3t} u(t-3)) &= e^{-3s} L(e^{-3(t+3)}) \\ &= e^{-3s} e^{-9} L(e^{-3t}) \\ &= \frac{e^{-3s-9}}{s+3} // \end{aligned}$$

$$\text{ii) } f(t) = L^{-1} \left[ \frac{\pi}{(s-3)^2 + \pi^2} \right]$$

$$\begin{aligned} f(t) &= L^{-1} \left[ \frac{\pi}{s^2 + \pi^2} \right]_{s \rightarrow s-3} \\ &= e^{3t} \sin \pi t \end{aligned}$$

4.b)  $\vec{F} = (x^2 - yz, y^2 - xz, z^2 - xy)$ , show  $\vec{F}$  is conservative field

To show conservative,  $\nabla \times \vec{F} = 0$

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2yz & y^2xz & z^2xy \end{vmatrix} = \vec{i}(-x+x) - \vec{j}(-y+y) + \vec{k}(-z+z) = 0 //$$

Thus,  $\vec{F}$  is conservative such that,

$$\vec{F} = \nabla \phi$$

$$\vec{F} \cdot d\vec{r} = \nabla \phi \cdot d\vec{r}$$

$$(x^2yz)dx + (y^2xz)dy + (z^2xy)dz = \left( \vec{i} \frac{\partial \phi}{\partial x} + \vec{j} \frac{\partial \phi}{\partial y} + \vec{k} \frac{\partial \phi}{\partial z} \right) \cdot \left( dx \vec{i} + dy \vec{j} + dz \vec{k} \right)$$

$$\text{or, } (x^2yz)dx + (y^2xz)dy + (z^2xy)dz = d\phi$$

Integrating both

$$\text{or, } \int (x^2yz)dx + \int (y^2xz)dy + \int (z^2xy)dz = \phi$$

$$\text{or, } \int x^2 dx + \int y^2 dy + \int z^2 dz - \int d(xyz) = \phi$$

$$\text{or, } \frac{x^3}{3} + \frac{y^3}{3} + \frac{z^3}{3} - xyz + c = \phi //$$

4.(c) Work done =  $\int \vec{F} \cdot d\vec{r} = ?$ ,  $\vec{F} = (3x^2, 2xz - y, z)$ , along  
 $x=t, y=\frac{t^2}{4}, z=\frac{3t^2}{8}$  from  $(0,0,0)$  to  $(2,1,3)$

$$\sum \phi_i$$

$$\vec{F} \cdot d\vec{r} = 3x^2 dx + (2xz - y)dy + z dz$$

$$x=t, \quad y=\frac{t^2}{4}, \quad z=\frac{3t^2}{8}$$

$$dx=dt, \quad dy=\frac{t}{2}dt, \quad dz=\frac{3t}{4}dt$$

When  $x=0, t=0$   
 $x=2, t=2$

$$\int_0^2 \vec{F} \cdot d\vec{r} = \int_0^2 \left( 3t^2 dt + \left( 2 \cdot t \cdot \frac{3t^2}{8} - \frac{t^2}{4} \right) \frac{t}{2} dt + \frac{3t^2}{8} \cdot \frac{3t}{4} dt \right)$$

$$= \int_0^2 \left( 3t^2 + \frac{3t^4}{8} - \frac{t^3}{8} + \frac{9t^3}{32} \right) dt$$

$$= \left[ t^3 + \frac{3}{8} \cdot \frac{t^5}{5} - \frac{t^4}{32} + \frac{9}{128} t^4 \right]_0^2$$

$$= \frac{441}{40} \text{ units}$$

5.a) Green theorem,  $\oint_C \vec{F} \cdot d\vec{r} = ?$

$$\vec{F} = (3x^2 + y)dx + 4y^2 dy, \quad C: (0,0), (1,0), (0,2)$$

(anti clockwise)

$$\frac{\partial F_1}{\partial y} = 1, \quad \frac{\partial F_2}{\partial x} = 0$$

Eqn of line from  $(0,2)$  to  $(1,0)$

$$y - 2 = \frac{0 - 2}{1 - 0} (x - 0)$$

$$y - 2 = -2x$$

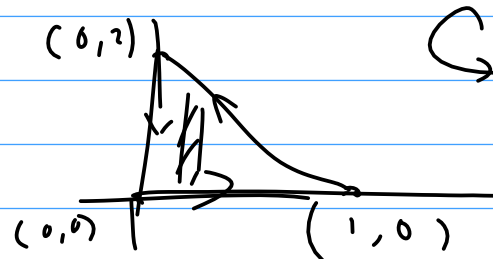
$$2x + y = 2$$

$$\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} = 0 - 1 = -1$$

Using Green's Theorem,

$$\iint_R \left( \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) dx dy = - \int_0^1 \int_0^{2-2x} dy dx = - \int_0^1 (2-2x) dx$$

$$\Rightarrow - \left[ 2x - x^2 \right]_0^1 = (2 - 1) = -1 //$$



$$R: 0 \leq y \leq 2-2x, \quad 0 \leq x \leq 1$$

5.b) Stoke's theorem:

let  $\vec{F}(x,y,z)$  be a continuously differentiable vector fn defined on an open surface  $S$  bounded by a curve  $C$

$$\text{Then, } \oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \vec{N} ds$$

$$\vec{F} = (y, xz^3, -2y^3) \quad ; \quad C: x^2 + y^2 = 4, \quad z = 3$$

$$= y\vec{i} + xz^3\vec{j} - 2y^3\vec{k}$$

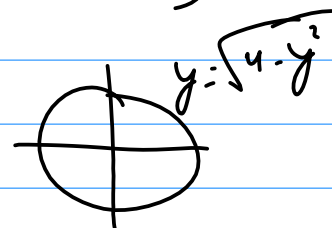
$$\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y & xz^3 & -2y^3 \end{vmatrix} = \vec{i}(-3zy^2 - 3xz^3) - \vec{j}(0 - 0) + \vec{k}(z^3 - 1)$$

$$= (-3zy^2 - 3xz^3)\vec{i} - 0\vec{j} + (z^3 - 1)\vec{k}$$

$$\vec{r} = x\vec{i} + y\vec{j} + 3\vec{k}$$

$$\vec{r}_x = \vec{i} + 0\vec{j} + 0\vec{k}, \quad \vec{r}_y = 0\vec{i} + \vec{j} + 0\vec{k}$$

$$\vec{N} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{vmatrix} = 0\vec{i} - \vec{j} + \vec{k} = \vec{k}$$



$$\iint (\nabla \cdot \vec{F}) \cdot \vec{n} \, dV = \iiint (z^3 - 1) \, dx \, dy \, dz = \iiint 26 \, dx \, dy \, dz$$

Changing to polar,  $0 \leq r \leq 2$ ,  $0 \leq \theta \leq 2\pi$

$$\int_0^{2\pi} \int_0^2 26r \, dr \, d\theta = 26 \int_0^{2\pi} \left( \frac{r^2}{2} \right)_0^2 d\theta$$

$$= 52 \times 2\pi = 104\pi //$$

OR

$$\iint_S \vec{F} \cdot \vec{n} \, dA = ? \quad , \quad \vec{F} = 18z\vec{i} - 12\vec{j} + 3y\vec{k} \quad , \quad \text{r: } 2x+3y+6z=12 \text{ in 1st octant}$$

Let  $\vec{r}$  be position vector,

$$\vec{r} = x\vec{i} + y\vec{j} + \frac{1}{6}(12-2x-3y)\vec{k} \quad , \quad z = \frac{(12-2x-3y)}{6}$$

$$\vec{r}_x = \vec{i} + 0\vec{j} - \frac{1}{3}\vec{k} \quad , \quad \vec{r}_y = 0\vec{i} + \vec{j} - \frac{1}{2}\vec{k}$$

$$\vec{N} = \vec{r}_x \times \vec{r}_y = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & -\frac{1}{3} \\ 0 & 1 & -\frac{1}{2} \end{vmatrix} = \vec{i}\left(\frac{1}{3}\right) - \vec{j}\left(-\frac{1}{2}\right) + \vec{k}(1)$$

$$= \frac{1}{3}\vec{i} + \frac{1}{2}\vec{j} + \vec{k}$$

$$\vec{F} = 3(12-2x-3y)\vec{i} - 12\vec{j} + 3y\vec{k} \quad ; \quad \vec{F} \cdot \vec{N} = (12-2x-3y-6+3y) = 6-2x$$

Taking projection on xy plane,  $z=0$

$$y = \frac{12-2x}{3}$$

$$\text{Put } y=0, z=0 \quad ; \quad x=6$$

We have surface integral,

$$\iint_S \vec{F} \cdot \hat{N} \, dS = \int_0^6 \int_0^{\frac{12-2x}{3}} (6-2x) \, dy \, dx = \int_0^6 \left[ 6y - 2xy \right]_0^{\frac{12-2x}{3}} dx$$

$$\Rightarrow \int_0^6 \left[ 6\left(\frac{12-2x}{3}\right) - 2x\left(\frac{12-2x}{3}\right) \right] dx \Rightarrow \int_0^6 \left( 24 - 4x - 8x + \frac{4}{3}x^2 \right) dx$$

$$\Rightarrow \int_0^6 \left( 24 - 12x + \frac{4}{3}x^2 \right) dx = \left[ 24x - 6x^2 + \frac{4}{9}x^3 \right]_0^6 = 24 //$$

6.b) Find Fourier sine & cosine series of  $f(x)=x^2$ ; interval  $0 < x < L$

Sol<sup>n</sup> The Fourier cosine series in interval  $[-L, L]$

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{L}$$

$$a_0 = \frac{1}{L} \int_0^L x^2 \, dx \Rightarrow \frac{1}{L} \times \left[ \frac{x^3}{3} \right]_0^L \Rightarrow \frac{L^2}{3}$$

$$\begin{aligned}
 a_n &= \frac{2}{L} \int_0^L x^2 \cos \frac{n\pi x}{L} dx \\
 &= \frac{2}{L} \left[ x^2 \sin \frac{n\pi x}{L} \cdot \frac{L}{n\pi} - \int 2x \sin \frac{n\pi x}{L} \cdot \frac{L}{n\pi} dx \right] \\
 &= \frac{2}{L} \left[ \frac{Lx^2}{n\pi} \sin \frac{n\pi x}{L} - \left( -\frac{2Lx}{n^2\pi^2} \cos \frac{n\pi x}{L} + \frac{2L^2}{n^3\pi^3} \sin \frac{n\pi x}{L} \right) \right]_0^L \\
 &= \left[ \frac{2x^2}{n\pi} \sin \frac{n\pi x}{L} + \frac{4Lx}{n^2\pi^2} \cos \frac{n\pi x}{L} - \frac{4L^2}{n^3\pi^3} \sin \frac{n\pi x}{L} \right]_0^L \\
 &= \left[ 0 + \frac{4L^2}{n^2\pi^2} (-1)^n - 0 - (0 + 0 - 0) \right] \\
 &= \frac{4L^2}{n^2\pi^2} (-1)^n
 \end{aligned}$$

$\therefore$  Fourier cosine series is,

$$x^2 = \frac{L^2}{3} + \sum_{n=1}^{\infty} \frac{4L^2}{n^2\pi^2} (-1)^n \cos \frac{n\pi x}{L} //$$


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Fourier sine series is,

$$f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L}$$

$$\begin{aligned}
 b_n &= \frac{2}{L} \int_0^L x^2 \sin \frac{n\pi x}{L} dx \\
 &= \frac{2}{L} \left[ -\frac{Lx^2}{n\pi} \cos \frac{n\pi x}{L} + \int 2x \cos \frac{n\pi x}{L} \cdot \frac{L}{n\pi} dx \right] \\
 &= \frac{2}{L} \left[ -\frac{Lx^2}{n\pi} \cos \frac{n\pi x}{L} + \frac{2Lx}{n^2\pi^2} \sin \frac{n\pi x}{L} + \frac{2L^2}{n^3\pi^3} \cos \frac{n\pi x}{L} \right]_0^L \\
 &= \left[ -\frac{2x^2}{n\pi} \cos \frac{n\pi x}{L} + \frac{4Lx}{n^2\pi^2} \sin \frac{n\pi x}{L} + \frac{4L^2}{n^3\pi^3} \cos \frac{n\pi x}{L} \right]_0^L \\
 &= \left[ -\frac{2L^2}{n\pi} (-1)^n + 0 + \frac{4L^2}{n^2\pi^3} (-1)^n - \frac{4L^2}{n^3\pi^3} \right] \\
 &= \frac{4L^2}{n^3\pi^3} (-1)^n - \frac{2L^2}{n\pi} (-1)^n - \frac{4L^2}{n^3\pi^3} \\
 &= \frac{2L^2}{n^3\pi^3} \left[ (-1)^n (2 - n^2\pi^2) - 2 \right]
 \end{aligned}$$

$\therefore$  Fourier sine series is,

$$x^2 = \sum_{n=1}^{\infty} \frac{2L^2}{n^3\pi^3} \left[ (-1)^n (2 - n^2\pi^2) - 2 \right] \sin \frac{n\pi x}{L} //$$



7.a) general sol<sup>n</sup> of  $u_x + u_y = u$

$$u_x + u_y - u = 0 \quad \text{Comparing with } au_x + bu_y - c = 0$$

$$a = 1, \quad b = 1, \quad c = u$$

$$\frac{dx}{1} = \frac{dy}{1} = \frac{du}{u}$$

Taking 1<sup>st</sup> & 2<sup>nd</sup>,

$$dx = dy$$

$$y = x + A$$

Taking 1<sup>st</sup> & 3<sup>rd</sup>,

$$dx = \frac{du}{u}$$

$$\log u = x + C$$

$$u = e^{x+C}$$

$$u = e^x \cdot C$$

$$u = B e^x$$

A is constant wrt x,

Take B as function of A i.e.  $B = f(A)$

$$B = f(y-x)$$

$\therefore$  general sol<sup>n</sup> is,  $u = F(y-x) e^x //$

7.b) Derive one-dimensional traffic flow,

Let  $\phi(x,t)$  be any flux term passing through 'x' and time 't'

$u(x,t)$  be any density fn. Then  $\phi = uv$

$$\phi(x,t) = u(x,t) \cdot v(x,t)$$

By conservation law,

$$u_t(x,t) + \phi_v(x,t) = f(x,t)$$

Here,  $f(x,t) = 0$  (source term is zero)

$$u_t + \phi_x = 0$$

Using  $\phi = \phi(u)$  we get,

$$\phi_x = \frac{\partial \phi}{\partial u} \cdot \frac{\partial u}{\partial x} = \phi' \cdot u_x, \quad \text{So, } u_t + \phi' u_x = 0$$

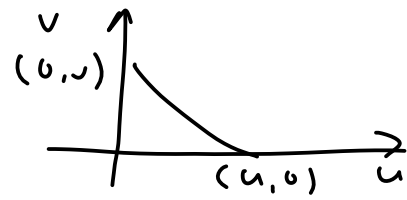
Next we find  $\phi'(u)$

Since speed of vehicle slows down due to dense traffic cond<sup>n</sup>. We can assume linear relationship bet<sup>n</sup> velocity & density of vehicle. To find the linear relationship. Let  $u_1$  and  $v_1$  be density & velocity of vehicle at some fixed pos<sup>n</sup> &

Eqn of st line passing through  $(u_1, v_1)$  and  $(0, v_0)$ . then

$$v - 0 = \frac{v_1 - 0}{0 - u_1} (u - u_1)$$

$$v = v_1 - \frac{u}{u_1} v_1$$



Put  $v$  in  $\phi = uv$

$$\phi = u \left( v_1 - \frac{u}{u_1} v_1 \right) = v_1 \left( u - \frac{u^2}{u_1} \right)$$

$$\phi' = v_1 \left( 1 - \frac{2u}{u_1} \right)$$

$$\therefore u_t + \phi' u_x = 0 \Rightarrow u_t + v_1 \left( 1 - \frac{2u}{u_1} \right) u_x = 0 //$$

7.c)  $x = t^3 + 1, y = t^2, z = 2t + 5,$   
 $v = 1, a = 1$  at  $t = 1$

SM<sup>r</sup> let  $\vec{r}$  be any position vector,

$$\vec{r} = x\vec{i} + y\vec{j} + z\vec{k} = (t^3 + 1)\vec{i} + t^2\vec{j} + (2t + 5)\vec{k}$$

Obs wrt  $t$ ,

$$\vec{v} = 3t^2\vec{i} + 2t\vec{j} + 2\vec{k}$$

$$\text{at } t = 1; \vec{v} = 3\vec{i} + 2\vec{j} + 2\vec{k} //$$

Obs. wrt  $t$ ,

$$\vec{a} = 6t\vec{i} + 2\vec{j} + 0\vec{k}$$

$$\text{at } t = 1; \vec{a} = 6\vec{i} + 2\vec{j} + 0\vec{k} //$$

Fin